

AI Lab # 8

This lab explores fuzzy set theory by implementing membership functions and fuzzy set operations to represent and reason with uncertainty and partial truth.

Artificial Intelligence (LAB)

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Date: 22 August, 2026

Today's Agenda

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What is Fuzzy Sets?

Fuzzy sets allow us to represent uncertainty or partial truth. Instead of just saying **yes or no**, we assign a **degree of membership** between 0 and 1.

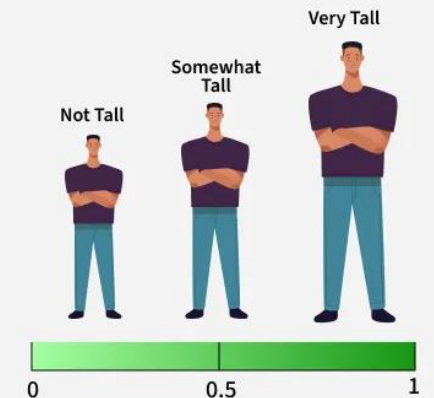
- A value of **0** means the element definitely does **not belong** to the set.
- A value of **1** means the element definitely **belongs** to the set.
- Values in between (like 0.4 or 0.7) represent **partial membership**.

For example, in a fuzzy set of "tall people":

- A person with a height of 6'2" might have a membership of **0.9** (very tall).
- A person with a height of 5'7" might have a membership of **0.5** (somewhat tall).
- A person with a height of 5'2" might have a membership of **0.2** (not tall).

Fuzzy logic

Fuzzy logic enables thinking in degrees, like describing someone as 'Somewhat Tall' or 'Very Tall' instead of just 'Tall' or 'Short.'



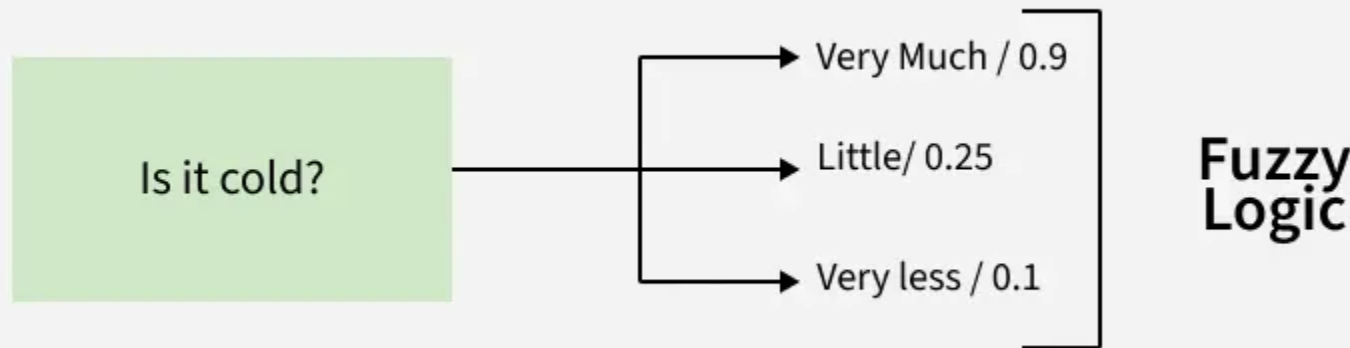
Real World Applications

- **Air Conditioners & Thermostats**

Instead of a rigid on/off at exactly 25°C, fuzzy sets define "warm," "hot," "cold" and adjust cooling power.

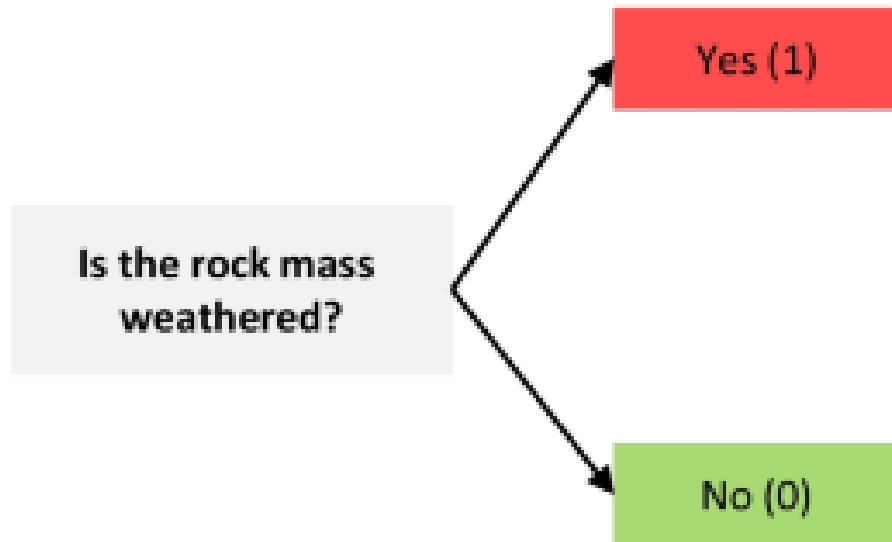
- **Anti-lock Braking Systems (ABS) in Cars**

Braking pressure is adjusted using fuzzy values like "slightly skidding," "moderately skidding," "heavily skidding" rather than a binary skid/no-skid decision.

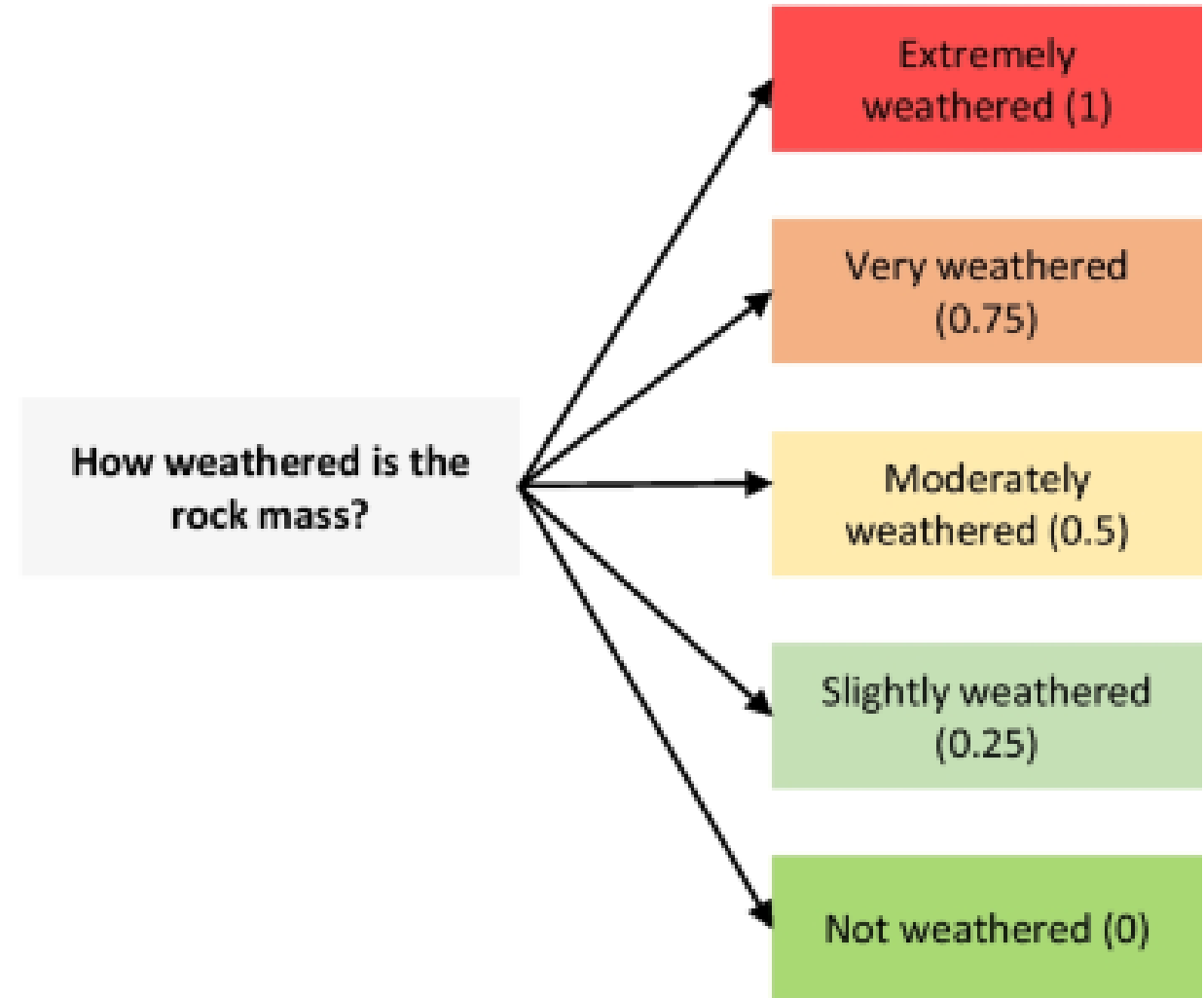


Crisp Sets vs Fuzzy Sets

CRISP SET



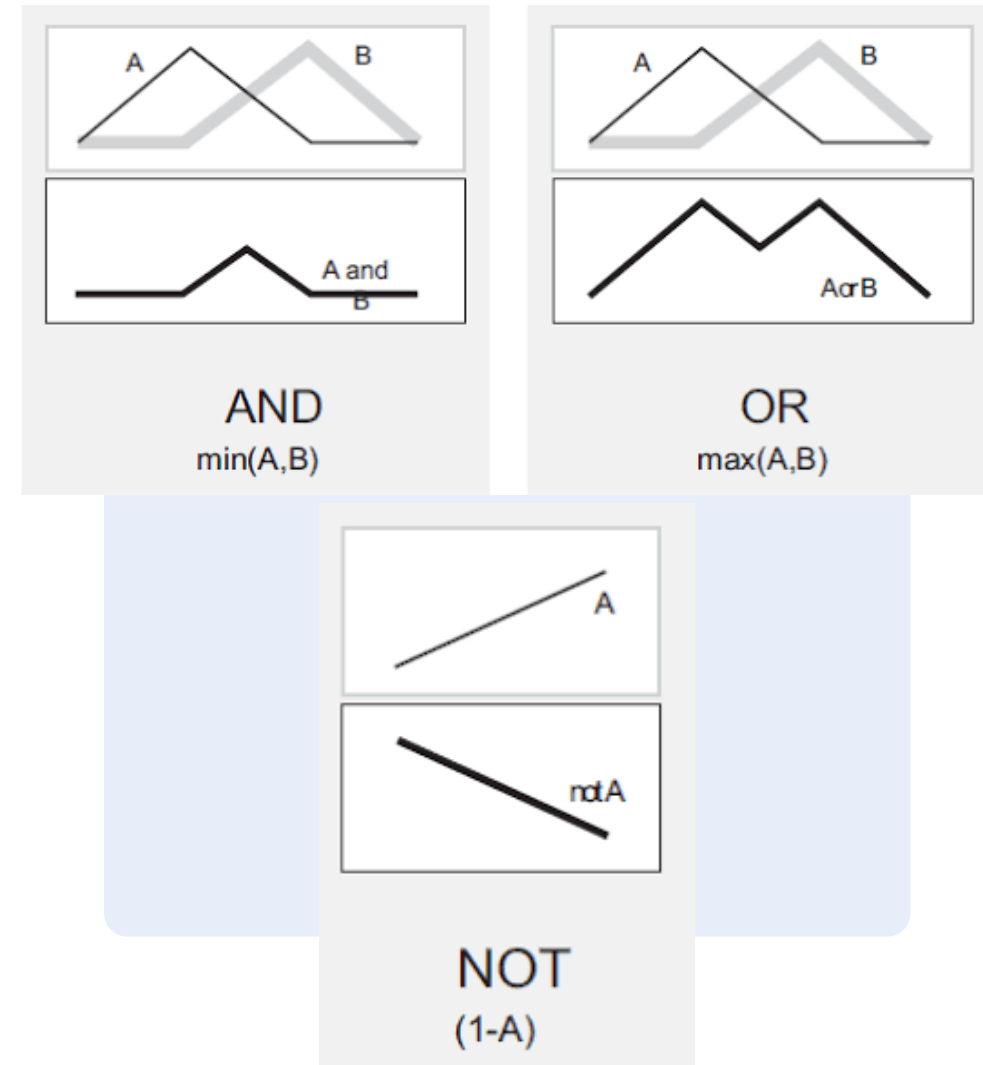
FUZZY SET



Fuzzy Sets Operations

Just like crisp sets, fuzzy sets also support operations such as **union**, **intersection**, and **complement**:

- **Union (OR)**: Combines two fuzzy sets by taking the maximum membership value for each element.
- **Intersection (AND)**: Combines two fuzzy sets by taking the minimum membership value for each element.
- **Complement (NOT)**: Inverts the membership value ($1 - \text{membership}$).



Union of Fuzzy Sets ($A \cup B$)

- Formula: $\mu_{A \cup B}(x) = \max(\mu_A(x), \mu_B(x))$
- An element's membership in the union is whichever set gives it the higher value
- **Example:**
- $A = \{0.2, 0.7, 1.0\}$, $B = \{0.5, 0.4, 0.9\}$
- $A \cup B = \{\max(0.2, 0.5), \max(0.7, 0.4), \max(1.0, 0.9)\}$
 $= \{0.5, 0.7, 1.0\}$

Code Snippet:

```
union = {}
for x in A:
    if A[x] > B[x]:
        union[x] = A[x]
    else:
        union[x] = B[x]
print("Union (A U B):", union)
```

Intersection of Fuzzy Sets ($A \cap B$)

- Formula: $\mu_{A \cap B}(x) = \min(\mu_A(x), \mu_B(x))$
- Takes the minimum membership value between the two sets
- **Example:**
- $A = \{0.2, 0.7, 1.0\}$, $B = \{0.5, 0.4, 0.9\}$
- $A \cap B = \{\min(0.2, 0.5), \min(0.7, 0.4), \min(1.0, 0.9)\}$
 $= \{0.2, 0.4, 0.9\}$

Code Snippet:

```
intersection = {}
for x in A:
    if A[x] < B[x]:
        intersection[x] = A[x]
    else:
        intersection[x] = B[x]
print("Intersection (A ∩ B):", intersection)
```


Complement of Fuzzy Sets (A')

- Formula: $\mu_{A'}(x) = 1 - \mu_A(x)$
- Inverts the membership value
- **Example:**
- $A = \{0.2, 0.7, 1.0\}$
- $A' = \{1-0.2, 1-0.7, 1-1.0\}$
 $= \{0.8, 0.3, 0.0\}$

Code Snippet:

```
complement = {}  
for x in A:  
    complement[x] = 1 - A[x]  
print("Complement of A:", complement)
```

Difference of Fuzzy Sets (A - B)

- Formula: $\mu_{A-B}(x) = \min(\mu_A(x), 1 - \mu_B(x))$
- Keep elements of A, reduced by how much they also belong to B
- Also called the relative complement of B in A

- **Example:**

- $A = \{0.2, 0.8, 1.0\}$, $B = \{0.5, 0.4, 0.7\}$
- $A - B = \{\min(0.2, 1-0.5), \min(0.8, 1-0.4), \min(1.0, 1-0.7)\}$

$$= \{\min(0.2, 0.5), \min(0.8, 0.6), \min(1.0, 0.3)\}$$

$$= \{0.2, 0.6, 0.3\}$$

Code Snippet:

```
difference = {}
```

```
for x in A:
```

```
    difference[x] = min(A[x], 1 - B[x])
```

Algebraic Sum of Fuzzy Sets ($A \oplus B$)

- Formula: $(A \oplus B)(x) = A(x) + B(x) - A(x) \cdot B(x)$
- Combines two fuzzy sets without exceeding a membership of 1
- **Example:**
- $A = \{0.7, 0.5, 0.2\}$, $B = \{0.6, 0.4, 0.8\}$
- $A \oplus B = \{0.7+0.6-(0.7 \times 0.6), 0.5+0.4-(0.5 \times 0.4), 0.2+0.8-(0.2 \times 0.8)\}$
 $= \{0.88, 0.70, 0.84\}$

Code Snippet:

```
algebraic_sum = {}
for x in A:
    algebraic_sum[x] = A[x] + B[x] - (A[x] *
B[x])
```

Algebraic Product of Fuzzy Sets ($A \otimes B$)

- Formula: $(A \otimes B)(x) = A(x) \cdot B(x)$
- Multiply the membership values for each element
- **Example:**
 - $A = \{0.2, 0.8, 1.0\}$, $B = \{0.5, 0.4, 0.7\}$
 - $A \otimes B = \{0.2 \times 0.5, 0.8 \times 0.4, 1.0 \times 0.7\}$
 $= \{0.1, 0.32, 0.7\}$

Code Snippet:

```
algebraic_product = {}
for x in A:
    algebraic_product[x] = A[x] * B[x]
```

Bounded Sum of Fuzzy Sets ($A \oplus B$)

- Formula: $(A \oplus B)(x) = \min(1, A(x) + B(x))$
- Add memberships; cap at 1 if it exceeds 1
- **Example:**
- $A = \{0.2, 0.8, 1.0\}$, $B = \{0.5, 0.4, 0.7\}$
- $A \oplus B = \{\min(1, 0.2+0.5), \min(1, 0.8+0.4), \min(1, 1.0+0.7)\}$
 $= \{\min(1, 0.7), \min(1, 1.2), \min(1, 1.7)\}$
 $= \{0.7, 1.0, 1.0\}$

Code Snippet:

```
bounded_sum = {}
for x in A:
    bounded_sum[x] = min(1, A[x] + B[x])
```

Bounded Difference of Fuzzy Sets ($A \ominus B$)

- Formula: $(A \ominus B)(x) = \max(0, A(x) - B(x))$
- Subtract B from A; floor at 0 if negative
- **Example:**
- $A = \{0.2, 0.8, 1.0\}$, $B = \{0.5, 0.4, 0.7\}$
- $A \ominus B = \{\max(0, 0.2-0.5), \max(0, 0.8-0.4), \max(0, 1.0-0.7)\}$
 $= \{\max(0, -0.3), \max(0, 0.4), \max(0, 0.3)\}$
 $= \{0.0, 0.4, 0.3\}$

Code Snippet:

```
bounded_difference = {}
for x in A:
    bounded_difference[x] = max(0, A[x] -
                                B[x])
```

Example of Fuzzy Sets

Now that we've covered Fuzzy Sets, let's move towards practical code-based example

Emotion in Classroom

Identify the current emotion of class fellow.



```
# Fuzzy set showing emotions of students
# Values between 0 and 1 tell how strongly they feel an emotion
emotions = {
    "Ali": {"Happy": 0.8, "Sad": 0.2, "Angry": 0.0, "Excited": 0.7},
    "Sara": {"Happy": 0.4, "Sad": 0.6, "Angry": 0.2, "Excited": 0.3},
    "Ahmed": {"Happy": 0.2, "Sad": 0.3, "Angry": 0.7, "Excited": 0.1},
    "Ayesha": {"Happy": 1.0, "Sad": 0.0, "Angry": 0.0, "Excited": 0.9}
}

# Detecting main emotion for each student
print("Detected Emotions in the Classroom:")
for student, emo_values in emotions.items():
    # Find the emotion with the highest value
    main_emotion = max(emo_values, key=emo_values.get)
    print(student, "is mostly", main_emotion)
```


Lab Task: Fuzzy Sets Operations

Use the given fuzzy sets: $A = \{x_1: 0.2, x_2: 0.8, x_3: 1.0\}$, $B = \{x_1: 0.5, x_2: 0.4, x_3: 0.7\}$, write code for the following operations

- Compute Algebraic Sum ($\oplus$)*
- Compute Algebraic Product ($\otimes$)*
- Compute Bounded Sum*
- Compute Bounded Difference*
- Make a table comparing the results of Union, Intersection, Algebraic Sum, Algebraic Product, Bounded Sum, and Bounded Difference*

Thank You

Questions or discussions are welcomed!